Quiz summary

Quiz number 1

Reynolds nr: $\mathbf{Re = UL/v}$

Drag on cylinder: $\mathbf{D = 0.5\ C_D\ \rho\ u_0^2\ d}$

if $\mathbf{Re << 1: C_D \sim Re^{-1}}$

if $\mathbf{Re > 100: C_D \sim 1}$

Rayleigh nr: $\mathbf{Ra = g\ \alpha\ \Delta T\ d^3 / v\ \kappa}$ (Heat diff. if small, else convection and turbulence)

Vector rules: $\mathbf{a \cdot (b \times c) = c \cdot (a \times b) = b \cdot (c \times a)}$

$\mathbf{a \times (b \times c) = b(a \cdot c) - c(a \cdot b)}$

Tensor rules: $\mathbf{U : U^T = U_{xx}^2 + U_{xy}^2 + U_{yx}^2 + U_{yy}^2}$

Ex if $\mathbf{T = f\,\hat{\imath}\,\hat{\imath} + g\,\hat{\jmath}\,\hat{\jmath}} \rightarrow \mathbf{\nabla \times T = (\hat{\imath}\,\partial_x + \hat{\jmath}\,\partial_y) \times (f\,\hat{\imath}\,\hat{\imath} + g\,\hat{\jmath}\,\hat{\jmath}) = (\hat{\jmath} \times \hat{\imath})\,\hat{\imath}\,\partial_x f + (\hat{\imath} \times \hat{\jmath})\,\hat{\imath}\,\partial_y g}$

$\rightarrow \mathbf{\nabla \cdot T = (\hat{\imath}\,\partial_x + \hat{\jmath}\,\partial_y) \cdot (f\,\hat{\imath}\,\hat{\imath} + g\,\hat{\jmath}\,\hat{\jmath})}$

Gauss law: $\mathbf{\oint u \cdot ds = \int (\nabla \cdot u)\, dV}$

Stokes law: $\mathbf{\oint u \cdot dl = \int (\nabla \times u) \cdot dS}$

Partial integration: $\mathbf{\int f\, \partial g/\partial x\, dx = [fg] - \int g\, \partial f/\partial x\, dx}$

Polar coordinates: $\mathbf{\nabla = (\check{r}\,\partial_r + \hat{O}/r\,\partial_\Theta)}$

$\mathbf{\partial \check{r}/d\Theta = \hat{O}}$

$\mathbf{\partial \hat{O}/d\Theta = -\check{r}}$

Continuity eq: $\mathbf{D\rho/Dt = \partial\rho/\rho t - \nabla \cdot (\rho u).}$

Mach nr: **Ma = flow speed / sound speed**

Reynolds nr: $\mathbf{Re = UL/v}$ **(Inertia / viscous forces)**

Vorticity: $\mathbf{\omega = \nabla \times u}$ (where $\mathbf{\omega = 2\Omega}$)

Circulation $\mathbf{C = \int u \cdot dl \approx \omega_z A}$

Vorticity eq: $\mathbf{D\omega/Dt = \omega \cdot \nabla u + v \nabla^2 \omega}$

Ideal 2D flow: $\mathbf{\partial\Psi/\partial y = u,\ \partial\Psi/\partial x = -v \rightarrow \nabla^2\Psi = \omega}$

(Note vorticity conserved if 2D since $\mathbf{\omega_z \rightarrow \omega \cdot \nabla u = 0}$)

Ideal 3D flow: Kelvin's thm: Circulation is preserved

Quiz number 2

The Kolmogorov spectrum: Going from the incromp-NS, we multiply with **u**:
$\mathbf{d/dt\ (\oint |u|^2 / 2\, dV) + 0 = - 0 - 1/ \nu \int \nabla u : (\nabla u)^T dV}$ ← energy decreases monotonically

Dimensional analysis gives $E_k = C\, \varepsilon^{2/3} k^{-5/3}$, where $\varepsilon \sim E/t_L = U^3/L$

2D turbulence: no vortex stretching, but we have enstrophy.

Reynolds decomposition: $\mathbf{\langle ab\rangle = \langle a\rangle\langle b\rangle + \langle a'b'\rangle}$

this makes us rewrite $\mathbf{\partial\Theta/\partial t + u \cdot \nabla\Theta \approx 0 \Rightarrow \partial/\partial z\ (\langle w'\Theta'\rangle) = 0}$
mixing length hypothesis $w' \sim l/\tau$ & $\Theta' = -\delta z \partial\langle\Theta\rangle/\partial z$, and assume $k_m = l^2/\tau$
$\Rightarrow \mathbf{\partial\langle\Theta\rangle/\partial t + \langle u\rangle \cdot \nabla\langle\Theta\rangle = -k_m \nabla^2 \langle\Theta\rangle}$
"Turbulent eddies carry a property (like heat or momentum) over a distance."

Rotating Boussinesq equations: $\mathbf{u_{in} = u + \Omega \times r \Rightarrow \Phi = g_{in} r - \Omega^2 r_h^2/2,\ \nabla\Phi = g\hat{z}}$

Coriolis: $\mathbf{\Omega = (\hat{y} \cos\theta + \hat{z} \sin\theta)\Omega}$, assume ŷ-part is small

Numbers: $\mathbf{Ro = U/fL,\ \ Ek = \nu/fL^2}$

Geostrophic flow: Low Ro and Ek and assume $\rho'/\rho_0 = - T'/T_0$
$\Rightarrow \mathbf{\partial u_h/\partial z = g/fT_0\, \hat{z} \times \nabla T'}$

Pressure coordinates: along y=const. $\Rightarrow \mathbf{dp = (\partial p/\partial x)_{y,z} dx + (\partial p/\partial z)_{x,y} dz}$
$\Rightarrow \mathbf{\partial u_h/\partial z = -R/fp\, \hat{z} \times \nabla_p T}$

Ekman layer: Assume $\rho' = 0$, $\nu \neq 0$, and the pressure distribution corresponding to geostrophic flow. $\mathbf{u' = -u_0 \cos(z/h_E)\, e^{-z/h_E}}$, where $\mathbf{h_E = sqrt(2\nu/f)}$.

Ekman transport: Define the turbulent vertical flux $\mathbf{\tau(z)}$ of horizontal momentum
$\mathbf{\tau = -\rho_0 K_m \partial u / \partial z}$

More clouds along the coast since the land has higher K_m and forces uplift

Ageostrophic flow: $\mathbf{u = u_g + u_a + \hat{z}w}$, $\mathbf{|u_a| / |u_b| \sim \varepsilon}$ and $\mathbf{Ek \sim Ro \sim \varepsilon}$ Remove geostrophic
$\Rightarrow \mathbf{u_a = 1/f\, \hat{z} \times (\partial u_g/\partial t + u_g \cdot \nabla u_g - \nu\nabla^2 u_g + k u_g)}$
1: Isollabaric wind
2: If acc → flow to LP, if dacc → flow to HP

3: Friction drives flow to LP
4: Towards LP

Gradient wind balance: $|V| << fr$, $\mathbf{V^2/r + fV = 1/\rho_0\ \partial p_d / \partial r}$

Quiz number 3

Brunt-Väisälä frequency: Buoyancy and Newton II → $\mathbf{N^2 = - g/\rho_0\ \partial \rho / \partial z}$

Internal gravity waves:
1) Define equilibrium, which simplifies
2) Linearize equations: $\mathbf{(u, \rho', p') \sim \varepsilon}$
3) Apply Fourier ansatz: $\mathbf{\sim exp[\ i\ (k \cdot r - vt)\ ]} \rightarrow \mathbf{\partial/\partial t = -iv,\ \nabla = ik}$
4) Obtain: $\mathbf{v^2 = f^2m^2 + N^2\ (\ k^2 + l^2) / k^2 + l^2 + m^2}$

Note: $\mathbf{Ro \sim v/f \sim 1} \rightarrow \mathbf{v^2 >> N^2} \rightarrow \mathbf{H^2 << L^2} \rightarrow$ Hydrostatic

Flow over mountain: Assume $\mathbf{(\partial \Psi / \partial t)_{x=const} = \partial \Psi / \partial t + u_0 \partial \Psi / \partial x} \rightarrow \mathbf{v = v_i + u_0\, k}$

Froude number : $\mathbf{u_0 / N\ H = Fr}$, derived from $\mathbf{f_{potential} = -\ \rho_0\, N^2\ z}$

Potential vorticity: Ro<<1, $\mathbf{P = \nabla \rho_d \cdot (\ w + \hat{z}\ f\ ) \approx (\ \xi + f\ ) / h}$
proof: Conisder $w_i = w + f\hat{z}$ and $c = w_i \cdot ds$

Rossby waves:
Barotropic flow: **u** is independent of **z**

Barotropic vorticity eq: Using streamfunction $\mathbf{\psi = p_d / f\rho_0}$, beta plane $\mathbf{f = f_0 + \beta y}$ and $\mathbf{\zeta = \nabla^2 \psi}$.
$\rightarrow \mathbf{\partial \psi / \partial t\ \nabla^2\ \psi + \beta\ \partial \psi / \partial x + (J\ \psi,\ \nabla^2\ \psi) = 0}$

Dispersion relation: Assume $u = (u_0, 0)$, $\psi = \psi_0 + \psi'$ and $\psi' \sim exp[\ i\ (kx + ly - vt)\ ]$
$\rightarrow \mathbf{v = u_0\, k - \beta\ k / k^2 + l^2}$

Jonas lecture summary

Ⅰ**)** Varying density wants to relax, ideally warm air would flow from the equator to the pole in upper troposphere and cold air below. But Coriolis turns this.

Ⅱ**)** Usually large: $Re = UL / v$, Also usually large $Ra = g\alpha\Delta T^3 H / v\varkappa$.
Both measure forcing versus energy sinks.

Ⅳ**–**Ⅴ**)** Fluid equations for ideal gas: Continuity, momentum, and energy. **Compressible**.

$Ma = u/c$, if Ma→0 then we assume Boussinesq. Assume $\rho = \rho_0 + \rho^{\sim}$. Neglect $\rho^{\sim}$ in the inertia term but keep in gravity. Linearizing energy eq.

Thus, we obtain **Bossinesq:** $\nabla \cdot u = 0$, and $D\rho^{\sim} /Dt = \varkappa\nabla^2\rho^{\sim} - \alpha\rho_0/C_p\, J_{rad}$

Ⅵ**)** For $\rho^{\sim} = 0$, we are incompressible, and *Re* is the only thing that determines flow.

Ⅵ**–**Ⅶ**)** Define vorticity as $\omega = \nabla \times u$. Incompressible NS gives $D\omega/Dt = \omega\cdot\nabla u + v\nabla^2\omega$.
This leads to vortex stretching, and if $v=0$, Kelvin's circulation theorem says the conservation of circulation. $\oint u \cdot dl = \oint \omega \cdot ds$.

For 2D-flow, we set $u = z \times\nabla \Psi$, and $\omega=\hat{z}\nabla^2\Psi$ and $D\omega/Dt = \varkappa\nabla^2\omega$. I.e 2d flow is less turbulent. And if $v=0$, ω is conserved, and point vortexes with trapped fluid in seperatix.

Ⅸ**)** From NS we derive an energy budget as $d/dt \int |u|^2/v\, dV = - v \int \nabla u:(\nabla u)^T dV$. RHS is negative if Re>>1. Since dissipation can only happen at scale d^{-1} we get a cascade as
$Ek = C\, \varepsilon^{2/3}\, k^{-5/3}$ where c=1.5. Note that the rate of dissipation is only given by *u.*

Ⅹ**)** Using Reynold averaging it can be shown that the advection of potential temperature gives a vertical heat flux. Using the mixing length hypothesis, this can be estimated as a turbulent diffusivity.

ⅩⅠ**) Boussinesq equations for a rotating system** give,
$Du/Dt + f(y)\hat{z} \times u = -\nabla p_d/\rho_0 - \hat{z}g\rho^{\sim}/\rho_0 + v \nabla^2 u$, $\nabla \cdot u = 0$, and $D\rho^{\sim} /Dt = \varkappa\nabla^2\rho^{\sim}$
Where the centrifugal force is in the gravity. Also $Ro = u/fL$, $Ek = v/fL^2$.
Up in the atmosphere, these are small, which gives $u_h \approx u_g = 1/f\rho_0 \hat{z} \times\nabla p_d$
Also the pressure is hydrostatic $\partial p_d/\partial z - g\rho^{\sim} = g\rho_0 T^{\sim}/T_0$ which gives $\partial u/\partial z = g/_f T_0\, \hat{z} \times\nabla T^{\sim}$.

ⅩⅡ**)** if $\rho^{\sim}=0$ then u_g is independent of z and must be zero at all levels. Thus we have $u=u_g+u_a$. We get a 45deg turning, and u_a gives transport to lower pressure. The thickness of the layer is $h_E = sqrt(2v/f)$ and $u_a \sim e^{-z/hE}$.

XIII) We set u = $u_g + u_a + \hat{z}w$, where u_a/u_g~Ro, only u_a can force vertical motion since $\nabla \cdot u_g=0$.
$\Rightarrow \partial w/\partial z = -\nabla \cdot u_a$. For $Ro<<1$, using perturbation analysis gives a contribution to u_a as:

1) Dissipative terms $\rightarrow u_a$ is towards LP
2) $-\partial u_g/\partial t \rightarrow$ Isollabaric flow (Towards low pressure until geostropic balance is achieved)
3) $\nabla\ u_g \cdot u_g \rightarrow$
   a) Changes in the speed of u_g make u_a is towards LP, where u_g increases until u_g is in geostrophic balance
   b) Changes the direction of u_g changes the centripetal acceleration of u_g as $u_a \sim |u_g|^2$. That strengthens u_g in an anticyclone and weakens it in a cyclone. See drawing.

The last point can be generalized to the gradient wind balance, which is valid for large Ro: $|u|^2/R + f|u| = 1/\rho_0\ \partial p_d/\partial r$, thus even anticylones can have LP in the center

XIV) Fouriers ansatz is $\Psi \sim e^{(k \cdot r - v t)}$. $c_x = v/k$ and $c_g \cdot \hat{x} = \partial v/\partial k$. Energy is c_g and the wave vector can be negative.

XV) Internal gravity waves: Derived from Boussinesq by following steps:

1) Define equilibrium (stratified and at rest)
2) Linerize
3) Fouriers ansatz
4) Solve

$v^2 = [\ f^2m^2 + N^2\ (\ k^2 + l^2)\ ]\ /\ k^2 + l^2 + m^2 = f^2 + (N^2 - f^2)cos^2\varphi$, where φ is the angle between the horizontal plane and k. We find $f < v < N$. The vertical phase velocity has the opposite sign to the vertical group velocity.

XVI-XVII) Flow over mountains: Intrinsic frequency is $v = v_i + u_0 k$. Assuming f=0 and using $v=0$ (which is true for lee waves), we obtain: $m^2 = (N^2 - u_o^2k^2)\ (k^2+l^2)/u_o^2k^2$. If $N^2<u_o^2k^2$, then the waves are evanescent.
An analytic solution can be found by BC as $w(0) = u_0\,\partial h/\partial x$ and assuming $h=h_0 \cos(k_0x)$, showing that the phaselines lean backwards.
If N decreases with height, one can obtain a trapped region.
Linear wave theory is only valid if $Fr>1$, where $Fr = u/NH$,

XVIII) Potential vorticity: From the rotating Boussinesq equations with zero viscosity We can derive $DP/Dt = 0$, where $P = [\ f(y) \cdot \hat{z} + w\ ] \cdot \nabla\tilde{\rho}$.
If $\nabla\tilde{\rho}$ is approximately vertical, we have $\nabla\tilde{\rho} = \hat{z}\ \delta\tilde{\rho}\ /\ h \Rightarrow P = [\ f(y) + \zeta\ ]\ /\ h.$

XIX) Rossby waves: Assume beta plane and $Ro<<1 \Rightarrow u \approx u_g = \hat{z} \times \nabla\ \Psi$ and $\zeta = \nabla^2\ \Psi$, where $\Psi=p_d/\rho_0 f_0$. $\Rightarrow D/Dt\ [\ \beta y + \nabla^2\ \Psi\ ]\ /\ h\ = 0.$ Assuming barotropic and constant h we obtain the barotropic vorticity equation: $\partial\psi/\partial t\ \nabla^2\ \psi + \beta\ \partial\psi/\ \partial x + (J\ \psi,\ \nabla^2\ \psi) = 0$
If we assume $\psi = \psi_0 + \psi' \rightarrow v = u_0 k - \beta\ k\ /\ k^2 + l^2$.

ⅩⅩ**)** Topographically forced Rossby waves: Stationary Rossby waves $v=0 \Rightarrow \beta/u_0=k^2+l^2$. Topography can be included by setting $h=H-h_T$ in the relation for P conservation. Assuming that h_T causes perturbations around a mean flow $u\hat{x}$, and only keeping terms $\sim h_T$ gives the equation as $(\partial/\partial t + u\, \partial/\partial x)\, \nabla^2 \psi' + \beta\, \partial\psi'/\partial x = -f_0/H\, u\, \partial h_T/\partial x$.
If h_T is periodic, we get infinity at resonance; we introduce friction. This gives a good explanation of deviation from zonal flow and why many cyclones form on the US East Coast.

ⅩⅩⅠ**) Quasi geostrophy**: Starting from the rotating Boussinesq equations Assuming $Ro \sim \varepsilon$. The lowest order gives geostrophic balance. When taking the curl of this equation only ageostrophic part remains since $\nabla \cdot u_g = 0$. Moreover $\partial w/\partial z = -\nabla \cdot u_a$ appears, which represents the vortex streathing. w is connected to the density in the energy equation and ones obtains. D/Dt P = 0, P = $\beta y + \nabla_h^2 \psi + f_0\, \partial/\partial z\, (1/N^2\, \partial\psi/\partial z)$, $u_g = \hat{z} \times \nabla\, \Psi$.

This quasi geostrophic P is an approximation of the one before. All other variables can be obtained from P (The invertibility principle). Which gives rise to the Omega-equation. This explains why cold air gives subsidence.